

Problem: Prove by mathematical induction that $\forall n \in \mathbb{Z}^+ \cup \{0\}, 6|7^n - 1$

Solution: Proof [*By Mathematical Induction*]. Let the predicate $P(k)$ be the statement $6|7^n - 1$. We will prove that $P(n)$ is true $\forall n \in \mathbb{Z}^+ \cup \{0\}$.

Basis Step: Show that $P(0)$ is true. $P(0)$ is true because $6|7^0 - 1 = 6|1 - 1 = 6|0$

Inductive Step: Show that $\forall n \in \mathbb{Z}^+ \cup \{0\}, P(k) \rightarrow P(k+1)$. Let $k \in \mathbb{Z}^+ \cup \{0\}$ and suppose that $6|7^k - 1$ (inductive hypothesis.) Show that $6|7^{k+1} - 1$. By definition of divisibility, the inductive hypothesis is equivalent to $\exists r \in \mathbb{Z}, 7^k - 1 = 6r$

$$\begin{aligned} \text{Then } 7^{k+1} - 1 &= 7 * 7^k - 1 \\ &= 7 * 7^k - 1 \\ &= (6 + 1)7^k - 1 \\ &= 6 * 7^k + (7^k - 1) \\ &= 6 * 7^k + 6r \\ &= 6(7^k + r) \end{aligned}$$

Now $7^k + r \in \mathbb{Z}$ because the products and sums of integers are integers. Thus by divisibility, $6|7^{k+1} - 1$.

Problem: A sequence d_1, d_2, d_3 is defined by letting $d_1 = 2$ or $\forall k \in \mathbb{Z}, k \geq 2, d_k = \frac{d_{k-1}}{k}$. Prove that $\forall n \in \mathbb{Z}^+, d_n = \frac{2}{n!}$

Solution: Proof (By mathematical induction) According to the definition of $d_1, d_2, d_3, \dots$, we have $d_1 = 2$ or $\forall k \in \mathbb{Z}, k \geq 2, d_k = \frac{d_{k-1}}{k}$. Let the predicate be the equality $d_n = \frac{2}{n!}$. We will prove that $P(n)$ is true $\forall n \in \mathbb{Z}^+$

Basis Step: Show that $P(1)$ is true. Left hand side of $P(1)$ is $d_1 = 2$, right hand side of $P(1)$ is $\frac{2}{1!} = 2$, so left hand side and the right hand side of $P(1)$ are equal, so $P(1)$ is true.

Inductive Step: Show $\forall k \in \mathbb{Z}^+, P(k) \rightarrow P(k+1)$. Let $k \in \mathbb{Z}^+$ and suppose that $d_k = \frac{2}{k!}$. We must show that $d_{k+1} = \frac{2}{(k+1)!}$

$$\begin{aligned} \text{Left hand side of } P(k+1) \text{ is } d_{k+1} &= \frac{d_k}{k+1} \\ &= \frac{\frac{2}{k!}}{k+1} \\ &= \frac{2}{k!(k+1)} \\ &= \frac{2}{(k+1)!} \end{aligned}$$

Problem: Prove by Mathematical Induction $\forall n \in \mathbb{Z}, n \geq 2, 5^n + 9 < 6^n$ **Answer:** Proof by Mathematical Induction. Let the predicate $P(n)$ be the statement $5^n + 9 < 6^n$. We will prove that $P(n)$ is true $\forall n \in \mathbb{Z}, n \geq 2$.

Basis Step: Show that $P(2)$ is true because the left hand side is $5^2 + 9 = 25 + 9 = 34$ and the right hand side is $6^2 = 36$ and $34 < 36$.

Inductive Step: Show that $\forall k \in \mathbb{Z}, k \geq 2, P(k) \rightarrow P(k+1)$. Let $k \in \mathbb{Z}, k \geq 2$ and suppose $5^k + 9 < 6^k$ (The inductive hypothesis) We must show that $5^{k+1} + 9 < 6^{k+1}$.

Multiply both the sides of the inductive hypothesis with 5 yields $5(5^k + 9) < 5 * 6^k$.

Note that $5^{k+1} + 9 < 5^{k+1} + 45 = 5(5^k + 9)$ and $5 * 6^k < 6^{k+1}$.

Thus: $5^k + 1 + 9 < 5(5^k + 9)$ and $5(5^k + 9) < 5 * 6^k$ and $5 * 6^k < 6 * 6^k = 6^{k+1}$

Thus, by transitivity, $5^{k+1} + 9 < 6^{k+1}$